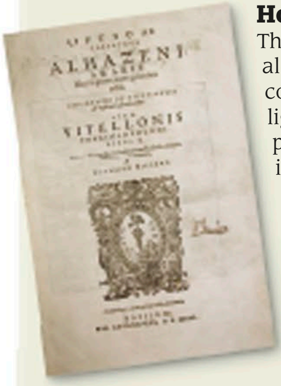
# Studying light

The scientific study of light and vision is called optics, and has been a subject of interest for hundreds of years. Several ancient civilizations, including the Ancient Greeks, experimented with light, and early Arab physicians had an advanced understanding of how the eye works. In the 1700s, an argument about whether light was in particle or wave form divided European scientists. We now know that light is a kind of electromagnetic radiation.



Front page of Latin version of the *Book of Optics*

### How we see

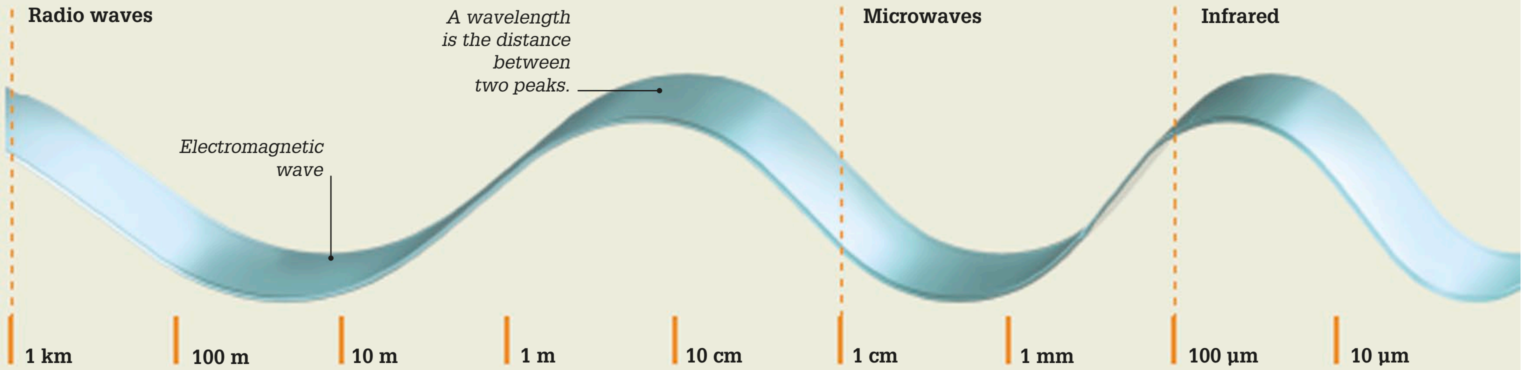
The *Book of Optics*, written by Arab scholar Ibn al-Haytham (Alhazen) between 1011 and 1021, correctly showed that vision occurs when rays of light enter the eye from outside. Until then, most people thought that visual rays extended from inside the eye outward.

Replica of Newton's color wheel

### Spinning colors

English scientist Sir Isaac Newton (see pp.88–89) used a color wheel to show that white light is made up of bands of color. When he spun the wheel, the colors—red, orange, yellow, green, blue, and violet—seemed to merge into white. We now know that colors have different wavelengths.

> **"... whenever all those Rays with those their Colours are mix'd again, they reproduce the same white Light as before."**
>
> Sir Isaac Newton, *Opticks*, 1704




µm—micrometer
nm—nanometer

### Radio waves

Radio waves are used to send sound and television signals. Dish telescopes pick up radio waves from space.

### Microwaves

Microwaves, or short-wavelength radio waves, are used to heat food and transmit mobile phone signals.

## Key events

**750 BCE**

The oldest known lens, from the Assyrian city of Nimrud (in present-day Iraq), was made of rock crystal and may have been used as a magnifying glass.

**1267**

Oxford scholar Roger Bacon described the structure of the eye. He made a study of lenses, and said that rainbows were caused by the reflection of sunlight from individual raindrops.

**c 1286**

The first spectacles, or eyeglasses, were made in Italy. Early spectacles, used by monks and scholars for reading, were balanced on the nose.

14th-century spectacles

**1678**

Dutch physicist Christiaan Huygens suggested that light is made up of spherical waves that spread out as they travel.